جـــامعـــة الـشـــارقـــة
UNIVERSITY OF SHARJAH

College of Computing and Informatics
Department of Computer Science

Course Syllabus

Course Title
Operating Systems

Course Code
1501352

Fall Semester, 2023/2024

Course Syllabus

Course Title	Operating Systems							
Course Code	1501352							
Credit Hours	(3 - 0: 3)			**Contact Hours**		(3 - 0: 3)		
Course Category		Univ. Req.		Col. Req.	✓	Prog. Req.	Others	
Course Type	✓	Compulsory			Elective			
Pre-requisite(s)	Data Structures (1501215) or equivalent							
Co-requisite(s)	None							
Semester	Fall	**Year**	2023/2024					
Instructor Name	Dr. Osman Abul							
Office Location	M5-209A							
Tel. No.								
Email	oabul@sharjah.ac.ae							
Lecture Times								
Office Hours	TBA on the Blackboard							

Course Description (as in the catalogue):

This course covers various topics in the arena of operating systems starting with the basics of operating systems and computer architecture and moving on to other important topics that include the following. Operating systems design and structuring, single threaded and multithreaded processes, process scheduling, process synchronization, memory management techniques, such as paging and segmentation, memory management techniques using virtual memory, such as demand paging and page replacement algorithms, storage structure management, and disk scheduling algorithms.

Course Objectives/Goals:

The goals of this course are to enable students to:

- Explain what an OS is and how it interacts with processes.
- Explain the interrelationship between the OS and the hardware.
- Explain the concepts of concurrent programming exemplified in the synchronization of processes and in IPC.

- Comprehend the different components of a modern OS.
- Implement various OS algorithms using a programming language.
- Evaluate different algorithms, such as process scheduling, disk scheduling and page replacement.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Explain the internals of the machine and how the OS and the hardware cooperate to present a convenient and efficient system to the user.
2. Examine the different design problems related to OS.
3. Discuss OS concepts like resource management, processes and their interaction, scheduling, and file system.
4. Implement the major algorithms needed in modern OS such as algorithms related to process scheduling, disk scheduling and memory management.
5. Evaluate the main algorithms of process, memory, and disk management used in modern operating systems.

Alignment of Course Student Learning Outcomes to Computer Science Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	**Analyze**	- Explain the internals of the machine and how the OS and the hardware cooperate to present a convenient and efficient system to the user.
	Apply	- Examine the different design problems related to OS.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	**Design**	- Examine the different design problems related to OS.
	Implement	- Implement the major algorithms needed in modern OS such as algorithms related to process scheduling, disk scheduling and memory management.
	Evaluate	- Evaluate the main algorithms of process, memory, and disk management used in modern operating systems.

Computer Science Program SLOs		Course SLOs
Communicate effectively in a variety of professional contexts.		- Discuss OS concepts like resource management, processes and their interaction, scheduling, and file systems.
Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Apply computer science theory and software development fundamentals to produce computing-based solutions.	**Theory**	- Discuss OS concepts like resource management, processes and their interaction, scheduling, and file system.
	Development	- Evaluate the main algorithms of process, memory, and disk management used in modern operating systems.

Alignment of Course Student Learning Outcomes to IT Multimedia Program Student Learning Outcomes

IT Multimedia Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	**Analyze**	- Explain the internals of the machine and how the OS and the hardware cooperate to present a convenient and efficient system to the user.
	Apply	- Examine the different design problems related to OS.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	**Design**	- Examine the different design problems related to OS.
	Implement	- Implement the major algorithms needed in modern OS such as algorithms related to process scheduling, disk scheduling and memory management.
	Evaluate	- Evaluate the main algorithms of process, memory, and disk management used in modern operating systems.

IT Multimedia Program SLOs		Course SLOs
Communicate effectively in a variety of professional contexts.		- Discuss OS concepts like resource management, processes and their interaction, scheduling, and file system.
Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Apply computer science theory and software development fundamentals to produce computing-based solutions.	**Theory**	- Discuss OS concepts like resource management, processes and their interaction, scheduling, and file system.
	Development	- Evaluate the main algorithms of process, memory, and disk management used in modern operating systems.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	Course SLO
1.	Introduction: - definition, user and system view, history of operating systems, computer system organization and architecture, operating system operations, virtualization, kernel data structures, computing environments.	Ch 1	1
2.	Operating System Structures: - user and operating systems interfaces, system calls, operating system services, operating system design and implementation, operating system structuring, building and booting an operating system.	Ch 2	1, 2
3.	Processes: - process concept, process scheduling types, process state transitions, process control block, operations on processes, interprocess communication (IPC), IPC in shared-memory and message passing systems, communication in client-server systems.	Ch 3	3
4.	- IPC in shared-memory and message passing systems, communication in client-server systems. - Programming based Tutorial Session: demonstration of the creation of processes, zombies, orphan processes in Ubuntu using C++.	Ch 3	3
5.	CPU Scheduling:	Ch 5	3

Week	Topic	Comments*	Course SLO
	- CPU-bound and I/O-bound processes, preemptive and nonpreemptive scheduling, scheduling criteria, scheduling algorithms (such as FIFO, shortest job first, shortest remaining time first, priority, round robin, multilevel queues, multilevel feedback queues).		
6.	- scheduling algorithms (such as FIFO, shortest job first, shortest remaining time first, priority, round robin, multilevel queues, multilevel feedback queues).	Ch 5	4, 5
7.	Multithreading: - benefits of multithreading, process vs. thread, multicore programming, multithreading models, threading issues (such as, fork() and exec() system calls, signal handling, thread cancellation, thread local storage, scheduler activations).	Ch 4	2, 3
8.	Process Synchronization: - critical section problem using consumer-producer model, race condition, preemptive and nonpreemptive kernels, mutex locks usage.	- Midterm Exam - Ch 6	2, 3
9.	- mutex locks on single core and multicore systems. - Semaphores- their usage and implementation, problems with semaphores (such as deadlock and starvation), priority inversion.	Ch 6	2, 3
10.	Memory Management: - background, address binding, logical vs. physical address space, dynamic loading and linking, contiguous memory allocation, fragmentation and its solution,	Ch 9	2, 3
11.	- paging, logical to physical address translation in paging systems, translation lookaside buffer, segmentation, logical to physical address translation in segmentation system.	Ch 9	2, 3
12.	Virtual Memory: - motivation, demand paging, requirements of demand paging system, page fault, page replacement algorithms (such as, FIFO, optimal page replacement, least recently used, most frequently used, least frequently used), least recently used algorithm implementations (such as stack based, reference bits, and second chance algorithm).	Ch 10	4, 5
13.	- page replacement algorithms (such as, FIFO, optimal page replacement, least recently used, most frequently used, least frequently used), least recently used algorithm implementations (such as stack based, reference bits, and second chance algorithm).	Ch 10	4, 5
14.	Mass Storage Structure: - magnetic disks, disk structures, disk management, file allocation table, bootable disks, disk scheduling algorithms (such as, FIFO, shortest seek time first,	Ch 11	3, 4, 5

Week	Topic	Comments*	Course SLO
	SCAN, C-SCAN, LOOK, C-LOOK), selecting a disk scheduling algorithm.		
15.	File System: - files, files attributes, file types, file operations, file structures, file allocation table. [If time permits]	- Ch 13 - Revision	1, 3
16.	**Final Exam**		

* In the comments, you can add the relevant chapter or notes, etc.

Scheduling of laboratory and other non-lecture sessions, including online sessions, as appropriate (if applicable)

Week	Topic	Comments
	N/A	

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date*	Comments
Homework #1	Week #4	
Homework #2	Week #8	

*The dates are tentative.

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date*	Weight (%)
Quizzes/itbls	Biweekly	15
Assignment	Week 4, 8	15
Midterm Exam	Week 9	25
Final Exam	Week 16	45
Total		**100**

*The dates are tentative.

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS Competencies
Explain the internals of the machine and how the OS and the hardware cooperate to present a convenient and efficient system to the user.	- Lectures	- Quizzes/itbls - Exams	60% of student achieve 60% of the CLO	Digital literacy
Examine the different design problems related to OS.	- Lectures - Tutorials	- Assignments - Exams	60% of student achieve 60% of the CLO	Critical and analytical thinking

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS Competencies
Discuss OS concepts like resource management, processes and their interaction, and scheduling.	- Lectures	- Quizzes/itbls - Exams	60% of student achieve 60% of the CLO	Digital literacy
Implement the major algorithms needed in modern OS such as algorithms related to process scheduling, disk scheduling and memory management.	- Lectures - Tutorials	- Assignments - Exams - Quizzes/itbls	60% of student achieve 60% of the CLO	Problem solving
Evaluate the main algorithms of process, memory, and disk management used in modern operating systems.	- Lectures	- Exams - Assignments - Quizzes/itbls	60% of student achieve 60% of the CLO	Critical and analytical thinking

Teaching and Learning Resources:

Textbook(s):

Silberschatz, A., Galvin, P. and Gagne, G. (2018). *Operating Systems Concepts* (10th Edition). Wiley Publishers. ISBN: 9781119320913.

Recommended Readings:

There are many books on OS including the ones by Tanenbaum and W. Stallings which are excellent references.

Other Resources:

Computer Usage:

Computers with C++ Compiler and Ubuntu operating system.

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

There are several assignments and quizzes that cover all the topics of the course. Students may work in teams on some assignments, which are normally the programming assignments, but must obtain the instructor's permission first.

Notes:

Teaching Methodology:

A hybrid of traditional lectures for 20-30 minutes followed by short sessions of questions and answers where students are encouraged to answer each other's questions, if any. This enforces students' ownership of the material taught in class. In order to achieve this objective, students are divided into small teams to examine and solve exercise(s) or provide feedback on some open problems. Discussion follows up among different teams.

Homework Assignments:

Students may work individually or in teams, depending on the topic and the instructor's discretion, on developing computer programs or/and solving a set of problems (problem-based learning technique).